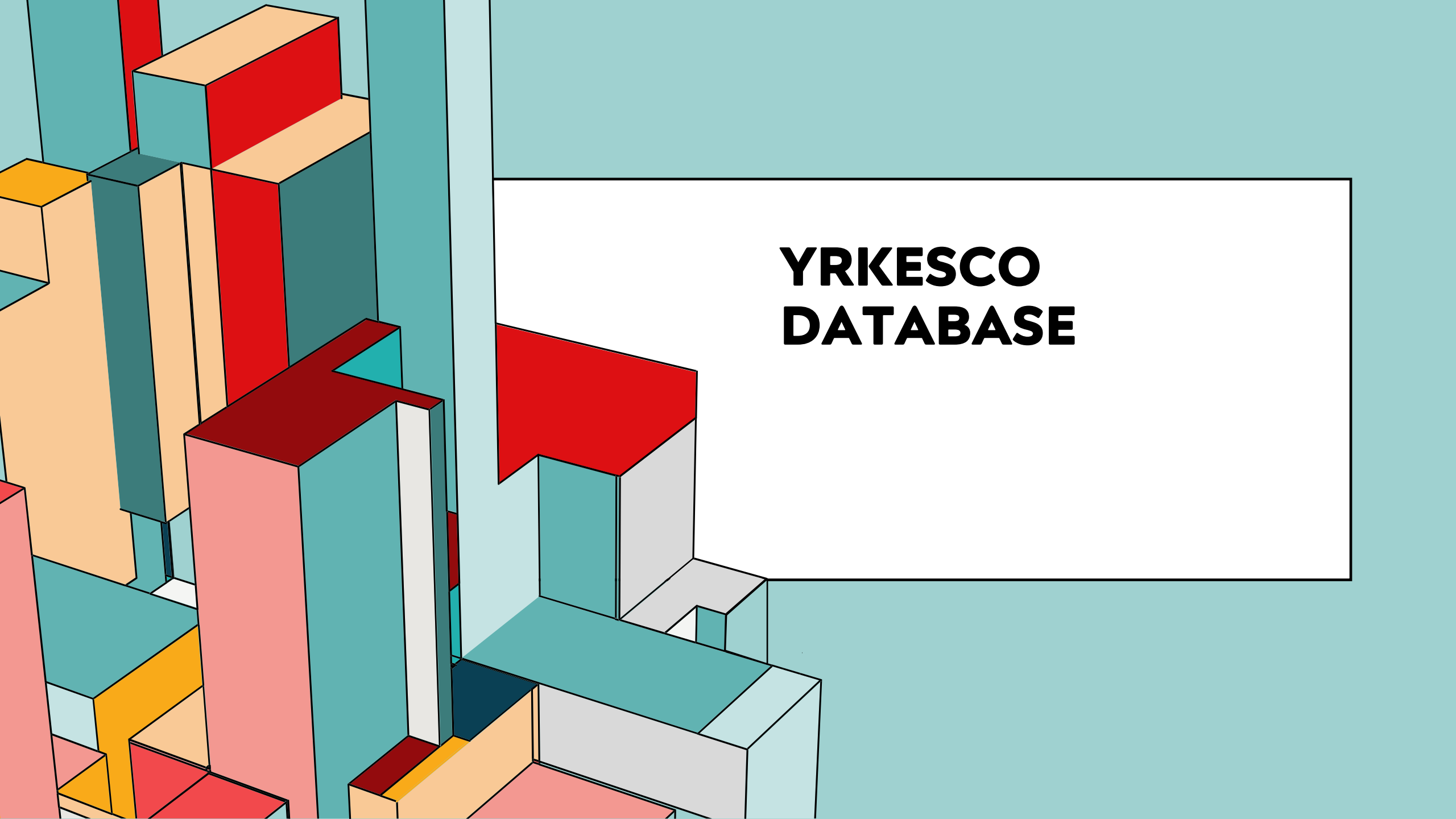
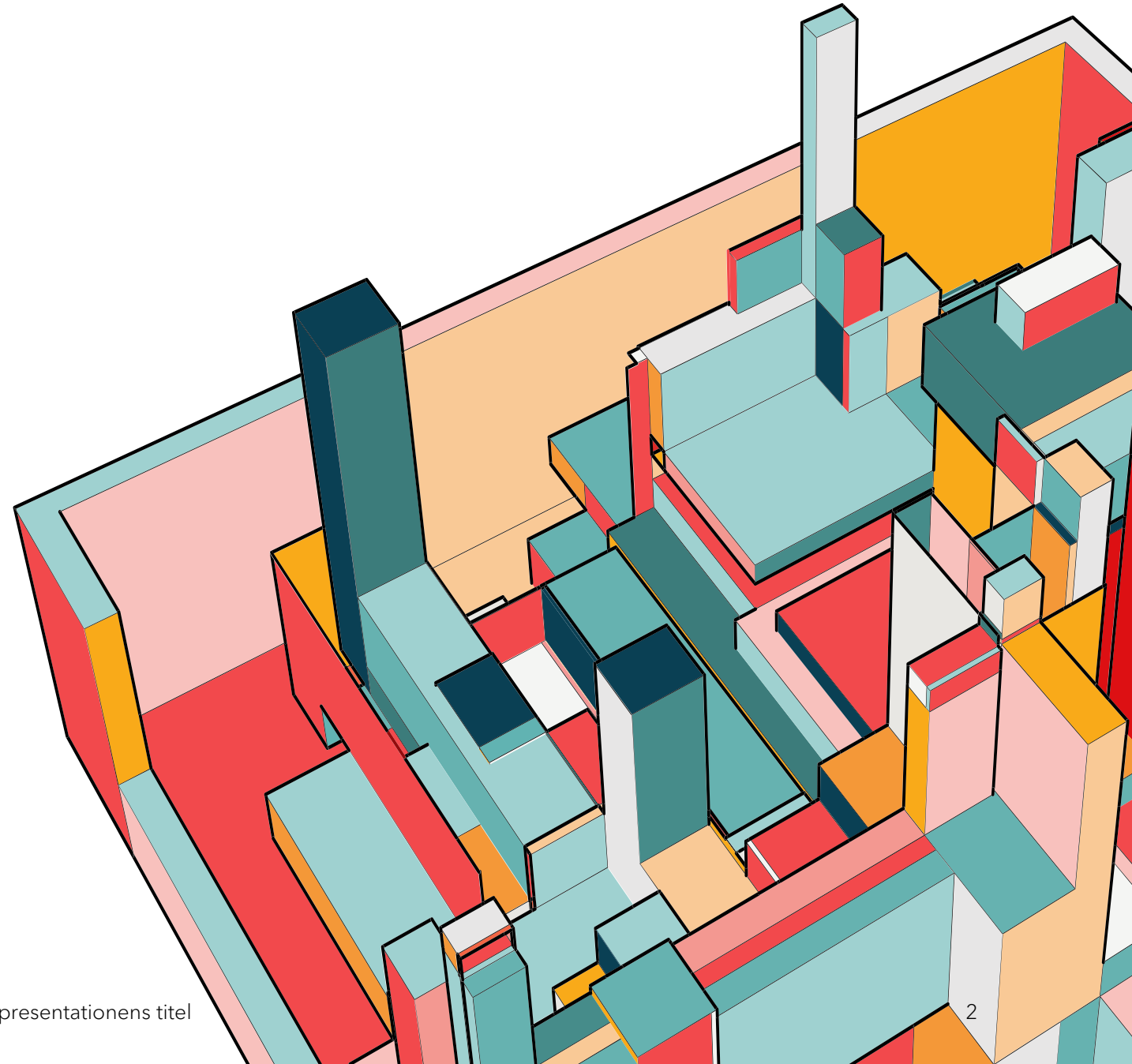


The background features a collection of 3D rectangular blocks in various colors including teal, orange, red, and pink, arranged in a staggered, isometric fashion. A large white rectangle with a black border is positioned on the right side of the image, containing the text.

YRKESCO DATABASE

WHAT AM I MAKING ?

- Conceptual_Model
- Logical_Model
- Physical_model



CONCEPTUAL MODEL OVERVIEW

People Domain (6 entities)

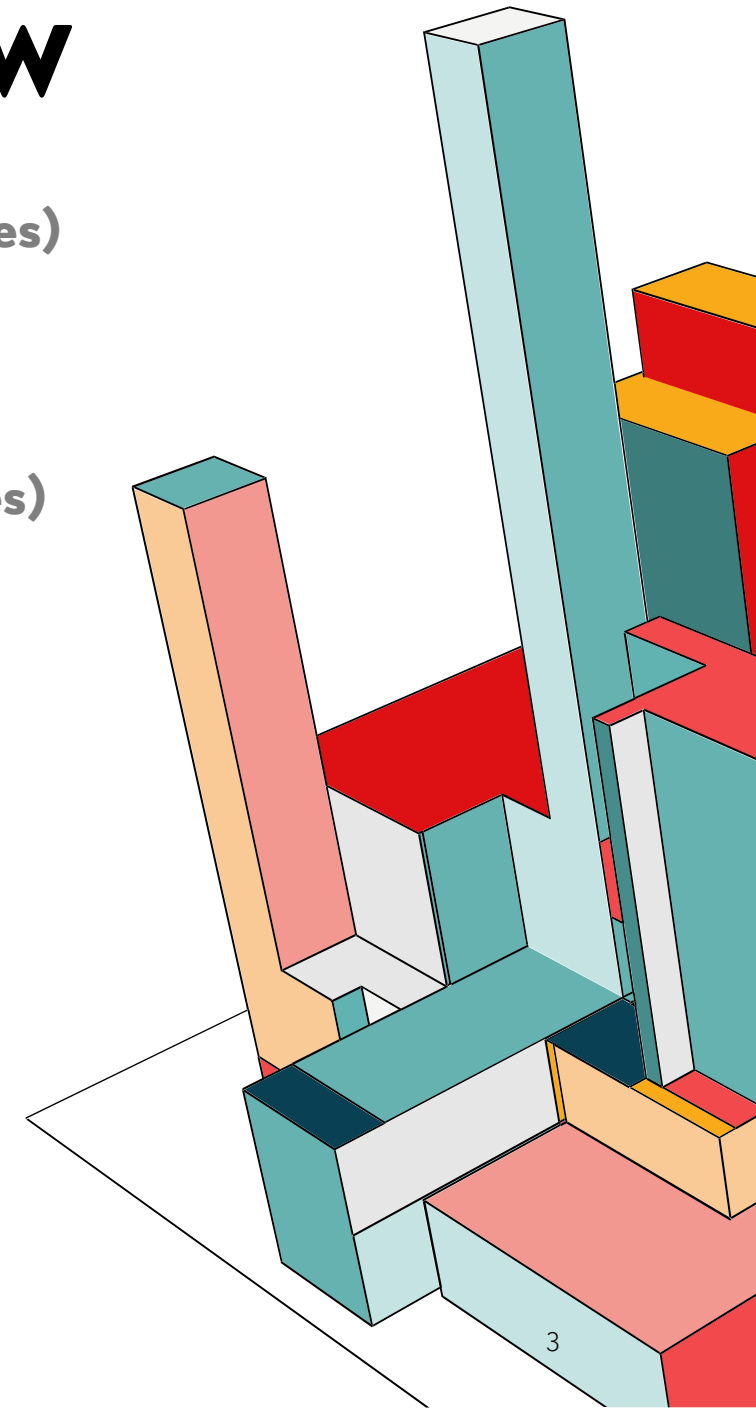
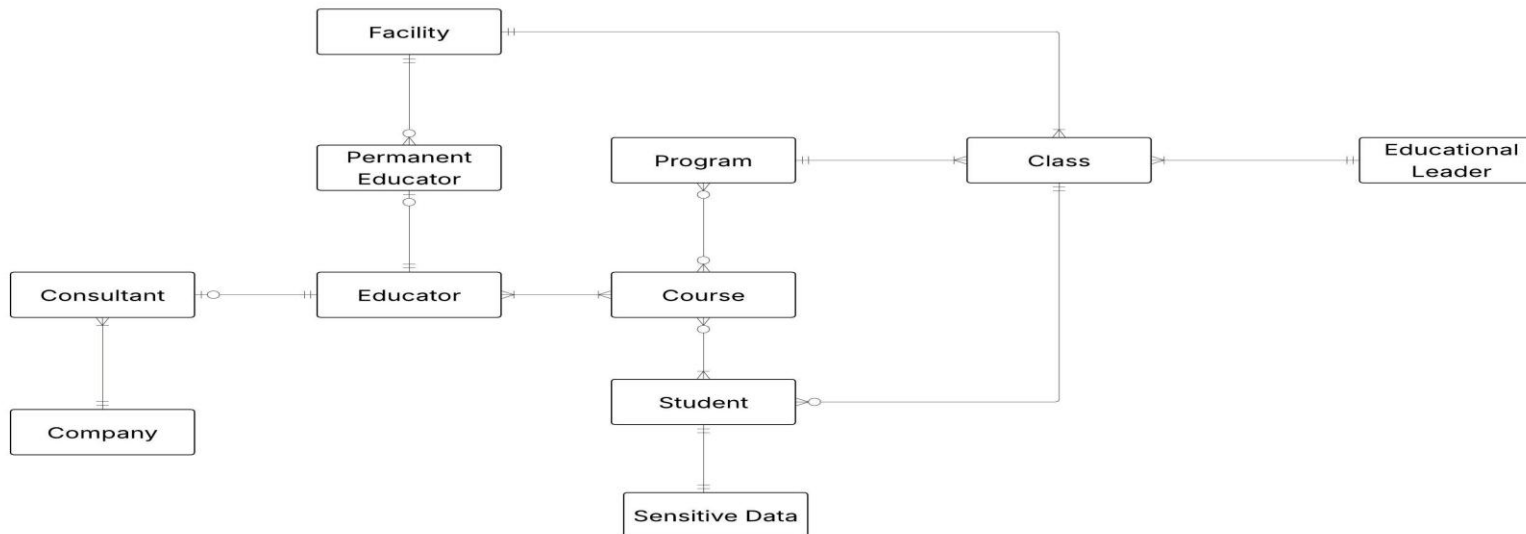
- STUDENT
- SENSITIVE_DATA
- EDUCATOR
- CONSULTANT
- PERMANENT_EDUCATOR
- EDUCATIONAL_LEADER

Organizational Domain (2 entities)

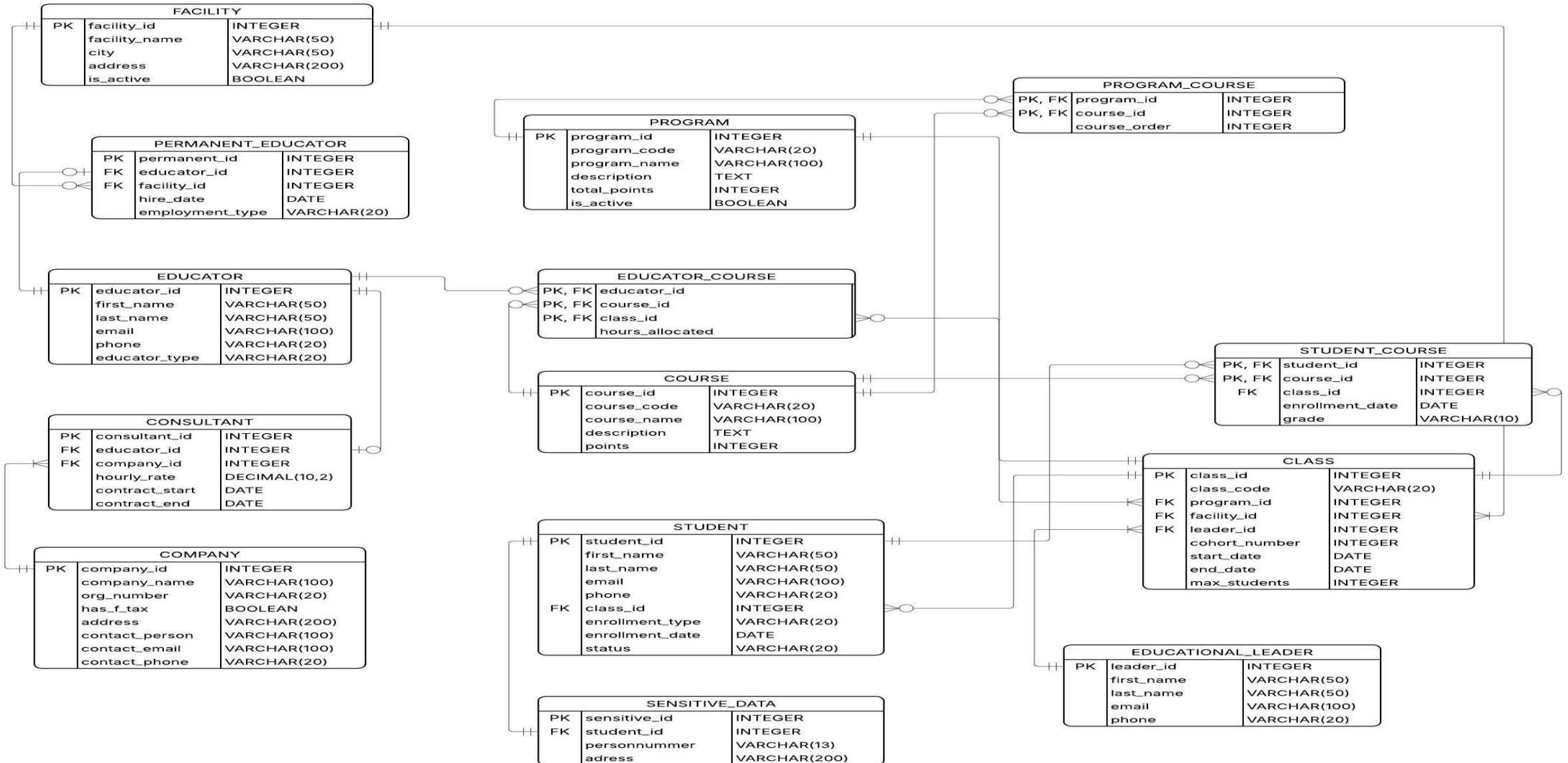
- Company
- Facility

Educational Structure (3 entities)

- Program
- Course
- Class



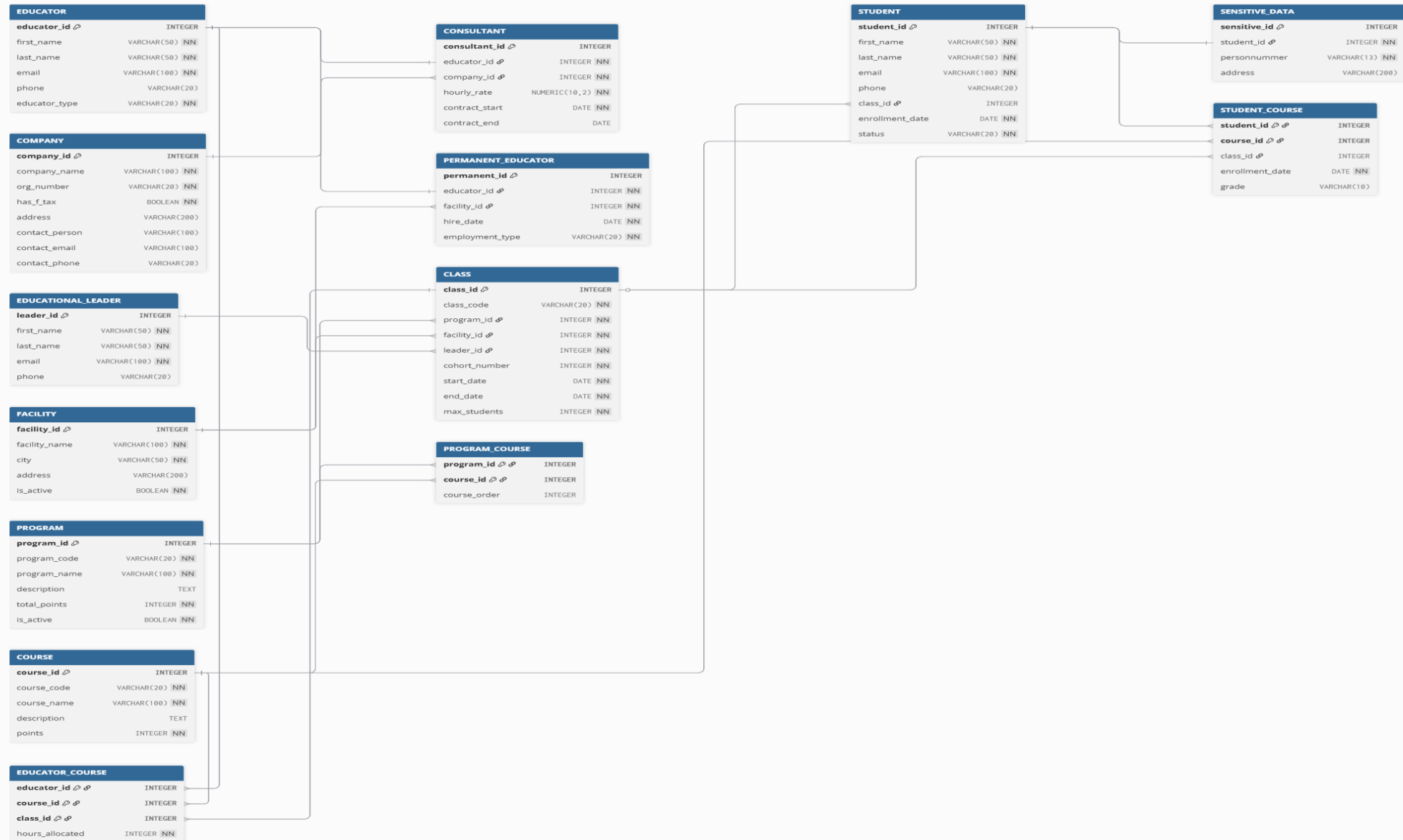
LOGICAL MODEL OVERVIEW



CONSTRAINTS = BUSINESS RULES

Constraint Type	Example	Business Rule
CHECK	cohort_number IN (1,2,3)	Exactly 3 cohorts
UNIQUE	(program_id, cohort_number)	One class per cohort
NOT NULL	enrollment_date NOT NULL	Required field
FK	class_id → CLASS	Referential integrity

PHYSICAL MODEL OVERVIEW



DOCKER SETUP

Why Docker ?

- Portable across environments
- Isolated from host system
- Easy setup & cleanup

docker-compose.yml:

```
services:  
  postgres:  
    image: postgres:latest  
    container_name: postgres
```

.env

```
POSTGRES_HOST="localhost"  
POSTGRES_USER="postgres"  
POSTGRES_PASSWORD="safe_password"  
POSTGRES_DB="yrkco_db"  
POSTGRES_PORT=5432
```

CREATE_TABLES.SQL

```
-- Create STUDENT table
CREATE TABLE
  IF NOT EXISTS STUDENT (
    student_id SERIAL PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    phone VARCHAR(20),
    class_id INTEGER,
    enrollment_date DATE NOT NULL DEFAULT CURRENT_DATE,
    status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
    FOREIGN KEY (class_id) REFERENCES CLASS (class_id)
  );
```


IMPLEMENTATION RESULTS

```
rkco_db=# \i sql/test_queries.sql
```

Class Code	Program	Facility	Educational Leader	Start Date	Max Students
WU23-K1-STO	Webbutvecklare	IT-Högskolan Stockholm	Maria Andersson	2024-01-15	30

1 row)

Class Code	Course Code	Course Name	Points
WU23-K1-STO	WEB101	Introduktion till Webbutveckling	50
WU23-K1-STO	WEB201	Backend-utveckling	50
WU23-K1-STO	WEB301	Frontend Frameworks	50

3 rows)

Class Code	Course Name	Educator	Type	Hours
WU23-K1-STO	Introduktion till Webbutveckling	Peter Johansson	PERMANENT	120
WU23-K1-STO	Backend-utveckling	Peter Johansson	PERMANENT	100
WU23-K1-STO	Frontend Frameworks	Mikael Larsson	CONSULTANT	80

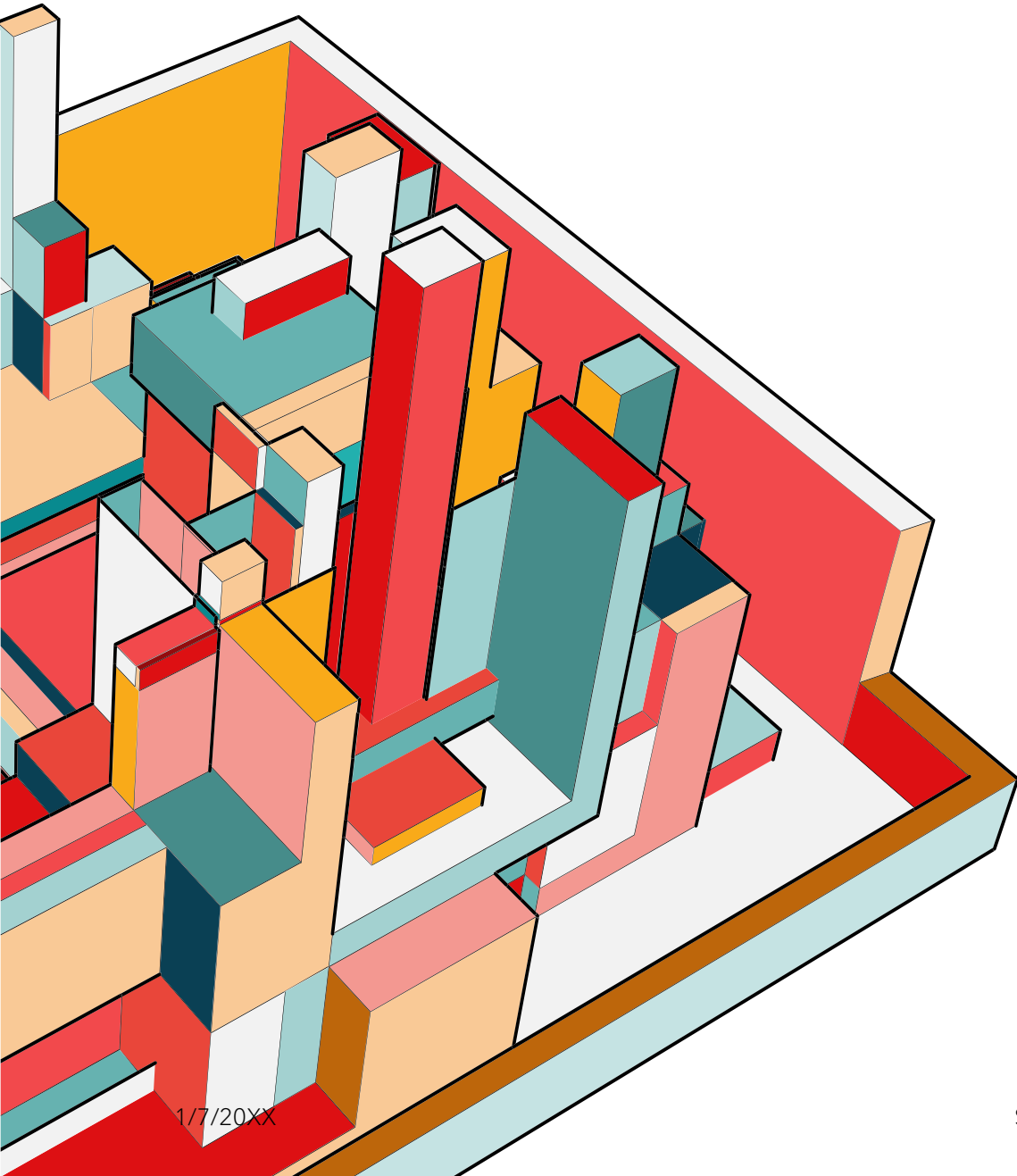
3 rows)

Class Code	Student Name	Email	Status
WU23-K1-STO	Julia Hedlund	julia.hedlund@student.se	ACTIVE
WU23-K1-STO	Emma Lundberg	emma.lundberg@student.se	ACTIVE
WU23-K1-STO	Oskar Sjöberg	oskar.sjoberg@student.se	ACTIVE

3 rows)

Consultant Name	Company	Hourly Rate	Contract Start
Mikael Larsson	Tech Consulting AB	950.00	2024-01-15
Karin Pettersson	Edukation Partner HB	850.00	2024-08-01

2 rows)



SUMMARY

Phase	Focus	Output
Conceptual	Business needs	11 entities + relationship
Logical	Structure + rules	PKs, FKs, constraints
Physical	Implementation	PostgreSQL tables
Setup	Deployment	Docker container + data

THANK YOU

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https://github.com/indiramahadiva/lab_1

_Indiramahadiva_DataModelling_DE25